

Regularization and Line Search

Saddle-Point Structure of KKT Systems

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Review: Inequality & Interior-Point

Previously:

$$\min_x f(x) \quad \text{s.t.} \quad g(x) \leq 0$$

KKT Conditions:

$$\nabla f(x) + \nabla g(x)^T \mu = 0$$

$$g(x) \leq 0$$

$$\mu \geq 0$$

$$\mu_i g_i(x) = 0$$

Today: Understanding the structure and solving them robustly.

Equality Constraints as Min-Max

Consider:

$$\min_x f(x) \quad \text{s.t.} \quad c(x) = 0$$

Equivalent to:

$$\min_x \max_{\lambda} f(x) + \lambda^T c(x)$$

Interpretation:

- If $c(x) \neq 0$, inner maximization blows up. - Only feasible x are stable.

Thus KKT point is a **saddle point**.

Saddle Point Structure

Lagrangian:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T c(x)$$

At optimum:

$$\min_x \max_{\lambda} \mathcal{L}(x, \lambda)$$

Second-order structure:

$$\nabla^2 \mathcal{L} = \begin{bmatrix} \nabla_{xx}^2 \mathcal{L} & \nabla c(x)^T \\ \nabla c(x) & 0 \end{bmatrix}$$

This matrix is:

quasi-definite

Positive definite in primal block, negative definite in dual block.

Why Regularization?

Newton step solves:

$$\nabla F(z) = 0$$

where

$$z = (x, \lambda)$$

But:

- Hessian may be indefinite - Overshoot possible - Newton not globally convergent

We need:

Line search

and sometimes:

Regularization

Root Finding + Line Search

Suppose solving:

$$F(z) = 0$$

Newton step:

$$z_{k+1} = z_k + \alpha \Delta z$$

But how choose α ?

Define scalar merit function:

$$\phi(z) = \frac{1}{2} \|F(z)\|^2$$

Now perform Armijo line search on ϕ .

Armijo Condition

Initialize:

$$\alpha = 1$$

While:

$$\phi(z + \alpha \Delta z) > \phi(z) + c\alpha \nabla \phi(z)^T \Delta z$$

reduce:

$$\alpha \leftarrow \beta \alpha$$

Typical values:

$$\beta \in (0, 1), \quad c \in (0, 10^{-4})$$

Ensures descent.

Merit Functions for Constrained Problems

We solve KKT system:

$$F(x, \lambda, \mu) = 0$$

Options for merit function:

1. Residual Norm

$$\phi = \|F(x, \lambda, \mu)\|^2$$

Simple and general.

2. Augmented Lagrangian Type

$$\phi(x) = f(x) + \frac{\rho}{2} \|c(x)\|^2$$

Better practical behavior.

Inequality Constraints as Min-Max

Consider:

$$\min_x f(x) \quad \text{s.t. } g(x) \leq 0$$

Equivalent idea:

$$\min_x \max_{\mu \geq 0} f(x) + \mu^T g(x)$$

If constraint violated:

$$g(x) > 0 \Rightarrow \mu \rightarrow +\infty$$

Again saddle structure.

Full KKT Newton System

Define:

$$F = \begin{bmatrix} \nabla_x \mathcal{L} \\ c(x) \\ S\mu - \tau e \end{bmatrix}$$

Newton system:

$$\nabla F \begin{bmatrix} \Delta x \\ \Delta \lambda \\ \Delta \mu \end{bmatrix} = -F$$

Jacobian structure:

$$\begin{bmatrix} \nabla_{xx}^2 \mathcal{L} & \nabla c(x)^T & \nabla g(x)^T \\ \nabla c(x) & 0 & 0 \\ \nabla g(x) & 0 & S \end{bmatrix}$$

This matrix must be quasi-definite.

Regularized KKT System

If indefinite or ill-conditioned:

Add regularization:

$$\nabla_{xx}^2 \mathcal{L} + \delta I$$

or

$$-\gamma I$$

on dual block.

Ensures:

- Better numerical stability - Better line-search behavior

Interpretation for Control

In robotics:

QP:

$$\min_x \frac{1}{2} x^T Q x + q^T x$$

$$A x = b$$

$$C x \leq d$$

Solver internally:

- Forms KKT system - Applies regularized Newton - Uses merit function + line search

This is why solvers like IPOPT are stable.

Key Takeaways

1. KKT conditions define a saddle point. 2. Newton method solves nonlinear KKT system. 3. Need line search for globalization. 4. Merit functions convert root finding into descent. 5. Regularization ensures quasi-definiteness. 6. This structure underlies modern MPC & whole-body control.